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An interpretable data-driven method for degradation prediction of proton exchange membrane fuel cells based on temporal fusion transformer and covariates

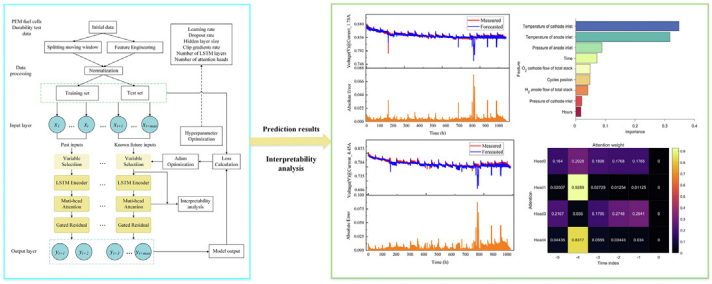
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HIGHLIGHTS

- Interpretable data-driven method for performance degradation prediction is proposed.
- Previously unknown and valuable information from experimental data is investigated.
- The root mean square error is 58% less than the attention-based gated recurrent unit.
- Feature importance and attention weights are obtained by interpretability analysis.

GRAPHICAL ABSTRACT



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ABSTRACT

The durability of proton exchange membrane (PEM) fuel cells is an obstacle to industrialization. PEM fuel cells need a robust incentive to improve their durability. Traditional data-driven methods combined with deep learning for degradation prediction methods lack interpretability, which prevents users from fully trusting the results and limits the application. Moreover, traditional data-driven methods not making full use of the valuable information in the experimental data. This study presents an easy to interpret data-driven method that improves the reliability of the model, the heterogeneity of input features, and the effects of significant covariates. The method's primary objective is to provide a long- and short-term degradation prediction of PEM fuel cells for a temporal fusion transformer. This method achieves high accuracy in predicting the degradation performance of PEM fuel cells with a root mean square error of 0.0067. The model's root means square error is reduced by 58% compared to the attention-based gated recurrent unit model. Based on the

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Temporal fusion transformer
Explainable artificial intelligence

interpretability analysis, the degradation performance directly correlates with power, the cathode inlet temperature, the anode inlet temperature, time, the cathode inlet pressure, and the cathode outlet pressure. The method can also be applied to fuel cell systems for optimizing control, management, and maintenance.

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Introduction

The over-consumption of fossil fuels has caused a serious energy crisis and environmental pollution, which has increased the global demand for clean and renewable energy sources [1]. Fuel cell vehicles (FCVs) appear to offer an alternative to conventional internal combustion engine vehicles [2]. FCVs based on PEM fuel cells have greater advantages in terms of energy efficiency, energy density, refueling time, and low-temperature performance. It has shown great promise for future transportation applications, especially in medium and long distances transportation [3]. However, the frequently varying operating conditions of FCVs (start, stop, idling, dynamic load) accelerate component deterioration of PEM fuel cells. The durability restricts the large-scale commercialization of current fuel cells [4].

Prognostic and health management (PHM) technology can improve fuel cell system reliability, availability, and durability by maintaining fuel cell systems primarily through fault diagnosis and prognosis [5]. The purpose of prognosis is to use historical data for predicting the remaining useful life (RUL) of fuel cells and assessing their health status. This allows strategic decisions to be made before failure occurs, effectively extending the lifetime of PEM fuel cells [6]. Therefore, an accurate prediction model is essential to the validity of prognosis. Besides, the prediction model helps to investigate the effect of operating parameters (such as temperature, pressure, and humidity) and conditions (such as start, stop, idling, load changing) on performance degradation and guides the monitoring of the degradation mechanisms related to PEM fuel cells. Various prognostic methods have been applied to the short-term and long-term degradation prediction of PEM fuel cells. These methods can be generally grouped into model-driven methods and data-driven methods.

Model-driven methods describe the degradation process from the perspective of physical phenomena. A mechanistic model of platinum catalyst degradation was developed to describe the interaction between the degradation rate of the electrochemical surface area and operating conditions [7]. Koltsova et al. [8] studied the effects of accelerated test cycles on the size of platinum nanoparticles in cathode catalysts based on a mathematical model developed for the degradation of platinum catalysts. The regularity of the electrochemical active surface area value over time can be employed to evaluate the lifetime of PEM fuel cells. However, model-based methods require a profound insight into the complex degradation mechanisms in the different components of PEM fuel cells. This is usually investigated using higher cost experiments. Moreover, these models are usually based on specific operating conditions.

In recent years, data-driven methods have been increasingly being deployed for predicting the degradation performance and remaining lifetime of PEM fuel cells based on the development of deep learning techniques. Deep learning applications have proven to be lower in cost and higher in accuracy compared to model-driven methods. Morando et al. [9] applied the echo state network (ESN) to the remaining useful life (RUL) estimation of the fuel cell stack. Mean cell voltage was used as network input and output. The average percentage error of the prediction results was less than 5%. However, the prognostic method used only past performance parameter voltages for prediction. Other important parameters such as current, temperature, pressure, and gas flow rates (i.e., gas mileage) also contain the degradation information of PEM fuel cells. Wu et al. [10]. Applied an improved prognostic model combining a hidden Semi-Mark model and an empirical model for predicting the RUL of Solid Oxide Fuel Cells, and the results showed that the proposed prediction model has superiority in prediction accuracy and prediction speed compared with the existing methods. Chen et al. [11] studied the effects of the parameter number on the results error in prediction. It was concluded that the prediction results considering five parameters are better than considering only two parameters. Increasing model dimensions help reflect nonlinear features more accurately. Ma et al. [12] proposed the prediction model based on the grid's long short-term memory network (LSTM). The pressure and temperature of reactants were fed into the model to strengthen the learning capability in time and space dimensions. This further improved the reliability of prediction results. Liu et al. [13] presented a prognostic method based on an adaptive neuro-fuzzy inference system (ANFIS) and an evolutionary algorithm. The evolutionary algorithm solved the parameter setting problems of ANFIS. Long-term prediction of stack voltage was implemented. Wang et al. [14] used a bidirectional LSTM model for long-term voltage prediction of fuel cell stacks. The random forest regression algorithm was chosen to select important variables. These variables included time, voltage, reactant temperatures at the inlet and outlet, and reactant flow rates at the outlet. Hua et al. [15] proposed multi-input and multi-output ESN for degradation prediction on the constant and dynamic operating conditions. They used voltage, current, temperature, and stack pressure as input parameters to the ESN. Pan et al. [16]. Used a model-based adaptive Kalman filtering and a data-driven NARX neural network to predict fuel cell degradation. The results indicated this method can provide accurate degradation predictions and the RUL. Mezzi et al. [17] developed an ESN-based model for predicting the fuel cells lifetime under a dynamic micro-cogeneration load profile. Zhang et al. [18] presented a multi-step strategy for degradation prediction. The multi-step prediction was

identified by its strengths over other strategies. The strength strategy also helped to achieve high accuracy in long-term forecasting and RUL estimation. Meraghni et al. [19]. Proposed a data-driven digital twin system with online prediction using a transfer learning model based on stacked denoising autoencoder. In which the proposed data-driven prediction method achieves high prediction accuracy. Zuo et al. [20] studied the long-term prediction performance of four types of recurrent neural networks: LSTM, gated recurrent unit (GRU), attention-based LSTM, and attention-based GRU under dynamic load cycles. The attention-based model has been used to help improve the accuracy of the model. Wang et al. [21] used the degradation behavior model and sliding window model identification methods to extract health indicators from fuel cells. A symbol-based LSTM was developed for predicting the degradation trend and the remaining lifetime of fuel cells. The proposed method could reflect the performance degradation characteristics under dynamic loads. Yue et al. [22]. Proposed an online fuel cell prediction method based on particle filtering method and developed a health management strategy system by combining this method, and the results showed that the designed system can effectively improve the durability of fuel cells. Li et al. [23] developed a prediction framework incorporating a bi-directional LSTM, a bi-directional GRU, and an echo ESN model, which achieves high accuracy in short-term degradation prediction and remaining useful life (RUL) prediction of PEM fuel cells with fewer data. Tang et al. [24] applied an end-to-end prediction algorithm based on a combination of migration learning and transformer neural networks to work with predicting the future decay trend of a PEM fuel cell stack. Benaggoune et al. [25] first developed a dilated convolutional neural network (Dil-CNN) for the degradation prediction of PEM fuel cells. The results showed that this method outperformed recurrent neural networks in short-term prediction. However, long-term prediction performance was not good enough. It had to utilize the trend of an autoregressive integrated moving average model as a guide for the Dil-CNN. The deterioration of the internal electrochemical reaction zone for PEM fuel cells was not considered by these methods. The degradation trend is tracked by building a black-box (BB) model that reflects the relationship between external parameters and time.

Lingenfelter and Khurshudov (2022) introduced the "BB Paradox" wherein models that are difficult to understand are usually accepted with fewer questions than those models that are easy to interpret. They found that black box machine learning models use big data analyses, which are so large and computationally complex they become "noisy and incomplete." Lingenfelter and Khurshudov explained this anomaly as a case where the developers of BB algorithms expect the development of more data to eventually verify their algorithm if they use that data to further train their model as they continue to interpret nonlinear relationships within the growing database [26]. This type of rationalization is said to excel in prediction but fall short in interpretability. However, most data-driven models are black-box. The structure of prediction models is governed by complex nonlinear interactions between internal parameters, which makes it difficult to describe how models generate predictions. Consequently, the results of these models are not trustworthy. Interpretability techniques can increase the credibility of the models and help model users to create rational strategies. Traditional data-driven methods fail to consider the heterogeneity of input features and the effects of significant covariates. By specifying the input features as a different type of input, the quality of the input data is improved to enhance the model performance with less heterogeneity of data sources. Most PEM fuel cell researchers focus on a few parameters but rarely on operating parameters like temperature, pressure, and humidity on performance degradation. These parameters affect the voltage or power decay of the fuel cell. Different parameters undoubtedly have different degrees of influence on performance degradation. Therefore, it is of great interest to develop interpretable prediction models that consider the heterogeneity of input features and the effects of the significant covariates for predicting the performance degradation of PEM fuel cells. The temporal fusion transformer (TFT) is a novel model for time series prediction [27]. It combines high-performance multivariate forecasting with interpretability. It has been successfully applied to predict several variables, such as traffic speed [28], meteorological data [29], and photovoltaic power generation [30].

In this study, the authors propose a novel degradation forecasting method based on a temporal fusion transformer and covariates. First, considering the heterogeneity of input covariates, 17 features derived from dynamic durability test data in the literature are grouped into time-varying past known covariates and future known covariates. Secondly, the degradation prediction model based on a temporal fusion transformer is developed by data preprocessing, a model topology definition, and hyperparameter optimization. Finally, the model results evaluation, comparison of different models results, and interpretability analysis are further investigated. The effects of different types of covariates on voltage degradation are investigated as well as the interpretability of the prediction model. The model has proven capable of further applications in fuel cell systems for optimal control, management and maintenance. The main innovations of this paper are.

- An interpretable data-driven method based on a temporal fusion transformer and covariates are first proposed for the degradation prediction of PEM fuel cells. Different data-driven models are compared with the interpretable data-driven method.
- Previously unknown and valuable information from experimental data is further investigated. Seventeen features are grouped into time-varying past known covariates and future known covariates according to the heterogeneity of the input covariates. Selected input features include time, dynamic cycle position, current, power, inlet and outlet temperatures, and pressure. These covariates have a significant effect on the performance degradation of PEM fuel cells.
- Explainable artificial intelligence is utilized to enhance the interpretability of prediction models. Based on the interpretability analysis, the importance of input covariates for performance degradation prediction is captured, and time-dependent attention weights are obtained. This analysis

provides guidance for the prognostics and health management (PHM) of fuel cells and gives direction for clarifying the cell degradation mechanism.

This paper is organized as follows. The theory of TFT and RNN is given in Section 2. Section 3 describes the dataset sources and data preprocessing, model development, and evaluation. Section 4 analyzes and discusses the model results, comparison, and interpretability. Finally, Section 5 presents the take aways to conclude the summary of this research.

Prediction methods

Different methods are proposed for predicting the degradation performance of PEM fuel cells such as recurrent neural networks. The model based on recurrent neural networks achieve a lot but have some limitations. This research aims to fully consider prediction accuracy and model interpretation to improve the credibility of results in the degradation prediction of PEM fuel cells. Besides, traditional data-driven methods do not consider the different types of input data. By specifying the input data as different data sources, the machine learner is able to deeply explores previously unknown valuable information in the dataset. In turn, the performance of the model is improved by reducing the heterogeneity of the input data. Most researchers rarely notice covariates even when they are part of their research experience, nor do they pay attention to known time-varying inputs (like temperature and pressure at inlets and outlets, cathode and anode flow, or relative humidity). These covariates are integrated into the model as a priori knowledge. The proposed models definitely include a temporal fusion encoder and temporal fusion decoder. The prediction of time dependencies is achieved through the long short-term memory and the multi-head attention network. The interpretable components are realized by the variable selection network and multi-head attention network.

Recurrent neural networks

A recurrent neural network (RNN) is proficient at learning sequential information in time series data. It has been widely used in speech recognition, text analysis, machine translation, sequence prediction, and other scenarios [31]. Fig. 1 shows the basic structure of an RNN. A simple RNN consists of an input layer, a hidden layer, and an output layer. The input layer is a given vector sequence, which is expressed as:

$$x = (\ldots, x_{t-1}, x_t, x_{t+1}, \ldots) \tag{1}$$

where x_t denotes input vectors at the moment t. The hidden layer enables the establishment of the relationship between past input vectors and output vectors. Meanwhile, the hidden layer can predict the future output vectors. The state of the hidden unit at any moment depends not only on input vectors at the current moment but also on the state of the hidden unit at earlier moments. It can be calculated as:

$$h_t = \tanh(W_{xh}x_t + W_{hh}h_{t-1} + b_h) \tag{2}$$

where tan h represents the activation function. W_{xh} denotes the weight matrix between the input layer and the hidden layer, W_{hh} and b_h indicates the weight matrices of hidden layers and the bias matrices, respectively. The weight matrix W between layers at different times is shared. In the output layer the vector sequence $y = (y_{t-1}, y_t, y_{t+1}, \ldots)$ is only associated with the hidden layer vectors:

$$y_t = \mathrm{softmax}(W_{hy}h_t + b_y) \tag{3}$$

where softmax $(W_{hy}\ h_t + b_y)$ is the normalized exponential function, W_{hy} represents the weight matrix between the hidden layer and the output layer, b_y denotes the bias vector. Additionally, the length of the output sequence is not necessarily equal to the length of the input sequence.

This structure enables RNN to process and model sequence data with any time step in theory. However, the problem of vanishing gradient occurs with the backpropagation of the gradient through time. This makes it difficult to handle longer sequence data.

Long and short-term memory network

LSTM is a special type of RNN. The hidden layer of LSTM introduces gate units and memory cells, which realize the function of memorizing long-term historical information [32]. Fig. 2 shows three gate units (the forget, input, and output gates) in an LSTM cell. At moment t, the input vector of the hidden layers is x_t and the vector of the memory cell is C_t. The

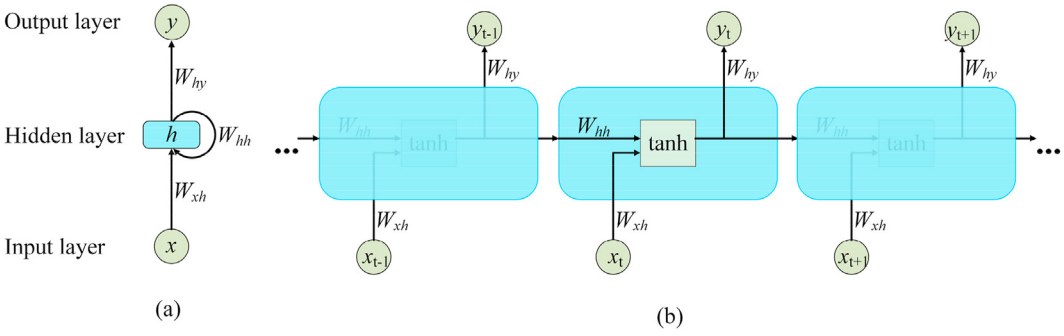


Fig. 1 – The basic structure of a recurrent neural network (RNN). (a) the folded structure (b) the unfolded structure at any time, t.

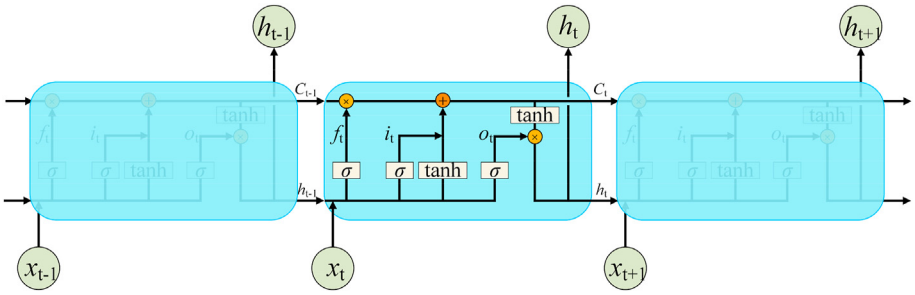


Fig. 2 – Three gate units which are forget, input, and output gate in an LSTM cell.

forget and input gates determine which information should be forgotten and remembered from the input layer and previous memory cell. The output gate determines the output information according to the input vectors. The following calculations represent the gate units and are expressed as:

$$f_t = \sigma(W_{xf}x_t + W_{hf}h_{t-1} + b_f) \quad (4)$$

$$i_t = \sigma(W_{xi}x_t + W_{hi}h_{t-1} + b_i) \quad (5)$$

$$o_t = \sigma(W_{xo}x_t + W_{ho}h_{t-1} + b_o) \quad (6)$$

where f_t, i_t and o_t represent the vector values of the forget, input and output gates, respectively. σ is the activation function; W_{xf}, W_{xi} and W_{xo} denote the weight matrices from the input layer to gate units; W_{hf}, W_{hi} and W_{ho} are the weight

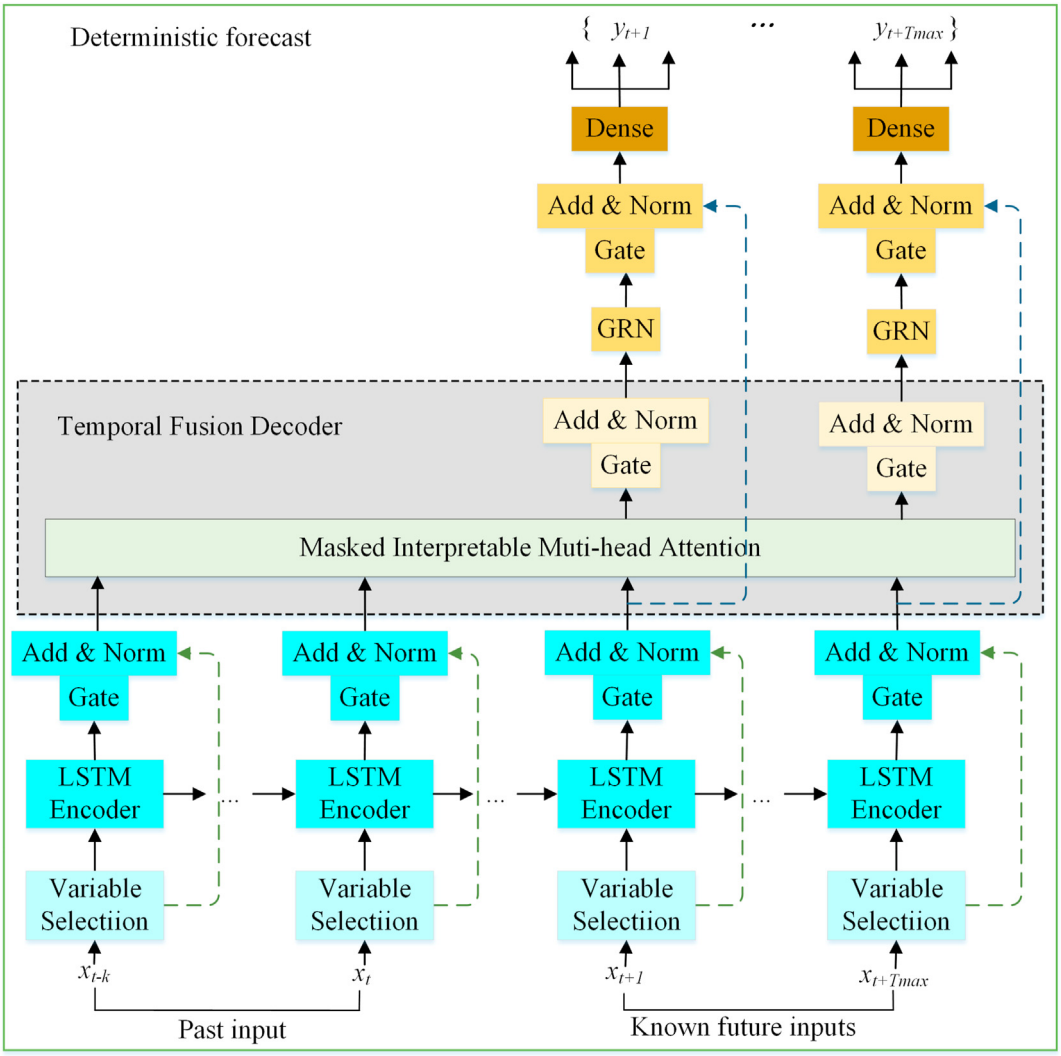


Fig. 3 – Model framework of temporal fusion transformer.

Table 1 – Parameters for degradation prediction.

Parameters	Units
Time	s
Current	A
Voltage	V
Power	W
Cycle position	
Pressure of anode inlet	Kpa
Pressure of cathode inlet	Kpa
Pressure of anode outlet	Kpa
Pressure of cathode outlet	Kpa
Temperature of anode inlet	°C
Temperature of cathode inlet	°C
Temperature of anode outlet	°C
Temperature of cathode outlet	°C
H2 anode flow of total stack	NLPM
O2 cathode flow of total stack	NLPM
Temperature of anode endplate	°C
Temperature of cathode endplate	°C
Temperature of anode dewpoint water	°C
Temperature of cathode dewpoint water	°C

matrices from the hidden layer to the gate units, and b is the bias vector of the different gates. The state of the memory cell is updated by the forget and input gates according to the following calculation:

$$C_t = f_t C_{t-1} + i_t \tanh(x_t W_{xc} + h_{t-1} W_{hc} + b_c) \quad (7)$$

where C_{t-1} denotes the vector value of the previous memory cell, tanh is the activation function, and h_{t-1} represents the output vector of a previous LSTM cell. W_{xc} and W_{hc} are the weight matrices from the input and hidden layer to memory cell, and b_c is the bias vector. Finally, the output of the LSTM cell to the output layer and the next cell is determined by the memory cell and the output gate cell together and is given as:

$$h_t = o_t \tanh(C_t) \quad (8)$$

Through the above three gate units, LSTM realizes the discarding, retaining and updating of information. The memory cell states in the hidden layer can flow over a long-time scale. This effectively solves the problem of vanishing gradients and better captures dependencies with large time steps in a time series.

Temporal fusion transformer

The temporal fusion transformer (TFT) is a transformer model for multi-step sequence prediction. Compared with traditional RNN, the TFT model is more robust based on its ability to design the network for different data sources. Furthermore, RNN fails to interpret the model. TFT improves the interpretability of prediction results by designing specific structures. Fig. 3 shows the model framework of TFT. The primary components of TFT are listed as follows.

1. Gated residual networks: They provide adaptive network complexity while ignoring any superfluous architectural components.
2. Variable selection networks: They sort out which input features are the most crucial at every point in time.
3. Interpretable multi-head attention networks: They interpret data based on past known time-varying input and future known input. Temporal processing may be used to learn both long- and short-term temporal correlations.

Gated residual network

Gated residual networks are used to reduce the effects of irrelevant output variables on the model and ensure the flow of valid information. Furthermore, it can be challenging to gauge how much nonlinear processing is necessary. Simpler models are probably more advantageous in some circumstances. Gated residual networks minimize the effect of noise and improve computational efficiency. The gated residual networks can be given as:

$$\mathrm{GRN}_\omega(\boldsymbol{a}) = \mathrm{LayerNorm}(\boldsymbol{a} + \mathrm{GLN}_\omega(\eta_1)) \quad (9)$$

$$\eta_1 = W_{1,\omega}\eta_1 + b_{1,\omega} \quad (10)$$

$$\eta_2 = \mathrm{ELU}(W_{1,\omega}a + W_{3,\omega}c + b_{2,\omega}) \quad (11)$$

$$\mathrm{GLN}_\omega(\gamma) = \sigma(W_{4,\omega}\gamma + b_{4,\omega}) \odot (W_{5,\omega}\gamma + b_{5,\omega}) \quad (12)$$

where a is the input vector. LayerNorm is layer normalization. $a + \mathrm{GLN}_\omega(\eta_1)$ can be considered as a linear contribution plus a nonlinear contribution. W and b indicate the weights and biases, respectively. ELU denotes the exponential linear unit activation function.

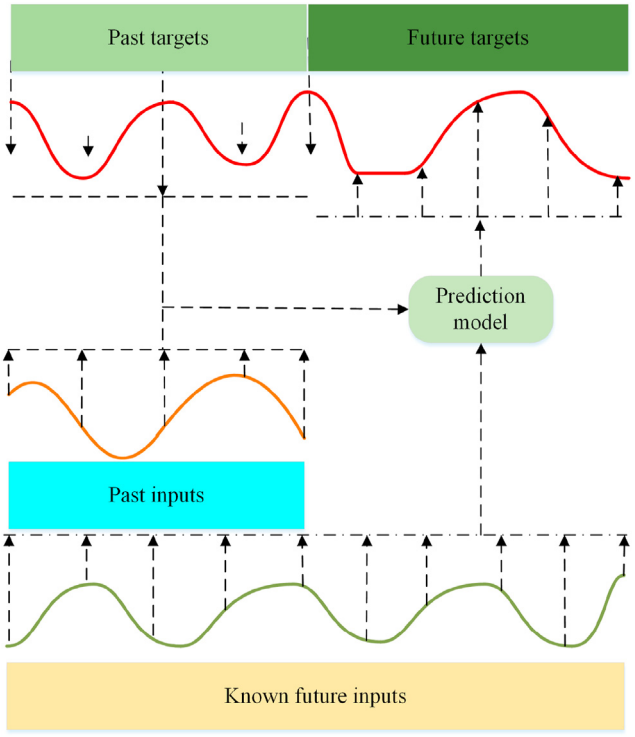


Fig. 4 – Illustration of forecasting with time-varying past known covariates and future known covariates.

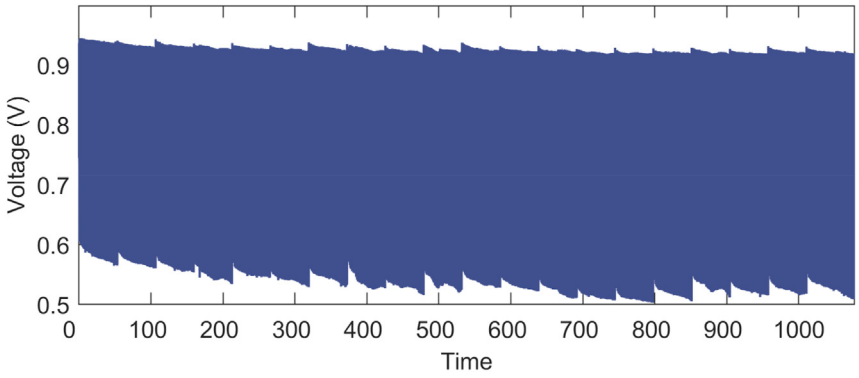


Fig. 5 – Final processing of the completed hours and the full voltage data.

Variable selection networks

Variable selection networks select the features that have the most significant effects on the output based on the input. Variable selection networks can be expressed as:

$$v_{x_t} = \mathrm{Softmax}(\mathrm{GRN}_{v_x}(\Xi_t)) \tag{13}$$

$$\tilde{\xi}_t^{(j)} = \mathrm{GRN}_{\tilde{\xi}_t^{(j)}}\left(\xi_t^{(j)}\right) \tag{14}$$

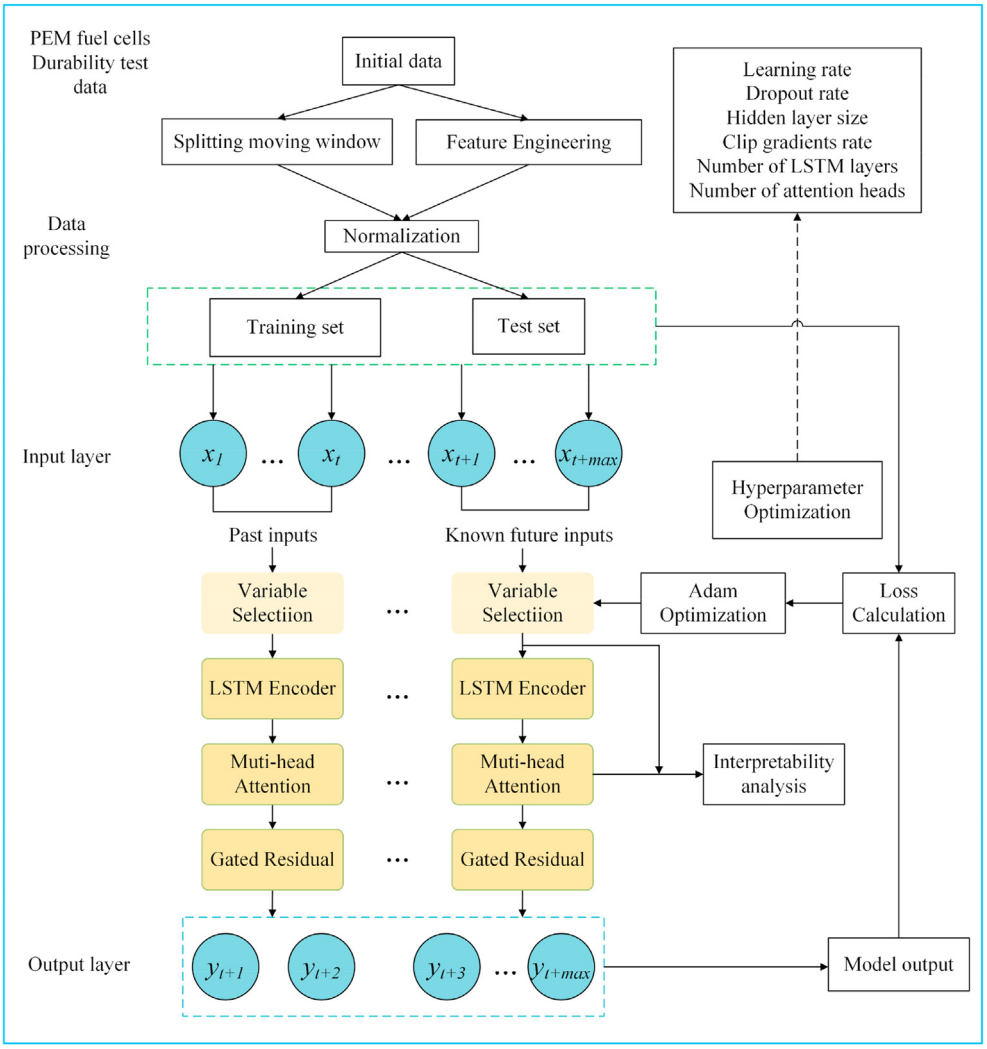


Fig. 6 – Training framework for predicting performance degradation of PEM fuel cells.

Table 2 – Statistical analysis of time-varying past known covariates of the model.

Variables	Count	Mean	Min	Max
Current	64085	8.540122	0.000000	35.541400
Power	64085	5.914640	0.000000	21.570000
Pressure of anode outlet	64085	110.0633	49.84314	116.99996
Pressure of cathode outlet	64085	107.7453	100.6776	110.81756
Temperature of anode outlet	64085	41.49630	26.58785	26.587851
Temperature of cathode outlet	64085	69.83488	68.06707	72.268265
Temperature of anode endplate	64085	83.14656	81.52598	84.851952
Temperature of anode dewpoint water	64085	54.95373	52.86728	56.424007
Temperature of cathode dewpoint water	64085	64.95541	59.39290	65.683617

$$\tilde{\xi} = \sum_{j=1}^{m_x} v_t^{(j)} \xi_t^{(j)} \tag{15}$$

where v_{x_t} indicates the matrix of variable selection weights. $\tilde{\xi}_t^{(j)}$ is the processed feature matrix for the feature j. $v_t^{(j)}$ denotes the j-th variable in the matrix.

Interpretable multi-head attention networks

Multi-head attention networks are proposed for improving the computing capacity of the attention mechanism [33]. Multi-head attention networks are used to learn long-term dependencies in serial variables and to improve the interpretability of the model. Interpretable multi-head attention can be expressed as:

$$\text{Atention}(Q, K, V) = A(Q, K)V = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_{attn}}}\right)V \tag{18}$$

$$\text{MultiHead}(Q, K, V) = [H_1, \ldots\ldots, H_{m_h}]W_H \tag{19}$$

$$\text{InterpretableMultiHead}(Q, K, V) = \tilde{H}W_H \tag{20}$$

$$= \frac{1}{H}\sum_{h=1}^{m_H} Attention\left(QW_Q^{(h)}, KW_K^{(h)}, VW_V^{(h)}\right) \tag{21}$$

Table 3 – Statistical analysis of time-varying future known covariates of the model.

Variables	Count	Mean	Min	Max
Hours	107660	504.601951	1	1009
Cycle position	107660	18	1	35
Pressure of anode inlet	107660	109.9000	49.37338	126.7975
Pressure of cathode inlet	107660	107.7661	99.25982	110.8175
Temperature of anode inlet	107660	69.96370	63.23562	73.72382
Temperature of cathode inlet	107660	69.83178	67.20259	72.26826
H2 flow of total stack	107660	0.146531	0.069000	0.497000
O2 flow of total stack	107660	0.610510	0.290000	2.071000

Table 4 – Optimal parameter values of the degradation prediction model.

Parameters	Min values	Max values	Optimal value
Optimizer			Adam
Learning rate	0.0001	0.01	0.0019
Hidden layer size	16	128	154
Dropout rate	0.01	0.3	0.2
Clip gradients rate	0.01	0.4	0.25
Batch size	16	512	256
Number of LSTM layers	1	10	1
Number of attention heads	1	16	4
Hidden continuous size	4	128	32

where A denotes the normalization function. $QW_Q^{(h)}, KW_K^{(h)}, VW_V^{(h)}$ are the head-specific weights for the keys, queries and values, respectively. m indicates the number of heads.

Interpretable degradation prediction of PEM fuel cells

In this section, the interpretable data-driven method is based on TFT for the degradation prediction of the PEM fuel cells. First, the dataset details are specified. Then, the specific data pre-processing and model development are explained. Finally, model evaluation is further investigated.

Introduction of the dataset

The dataset of this research is obtained from long-term dynamic durability test datasets [34]. It is carried out on a single PEM fuel cell for automotive applications with an active area of 25 cm^2. The dataset contains over 3.6 million data points and 17 features. The specific parameters used for the degradation prediction are listed in Table 1. In this study, the output voltage is identified as the forecast target. All other parameters are fed into the prediction model as covariates. These covariates, affecting the performance degradation, are grouped into time-varying past known covariates and future

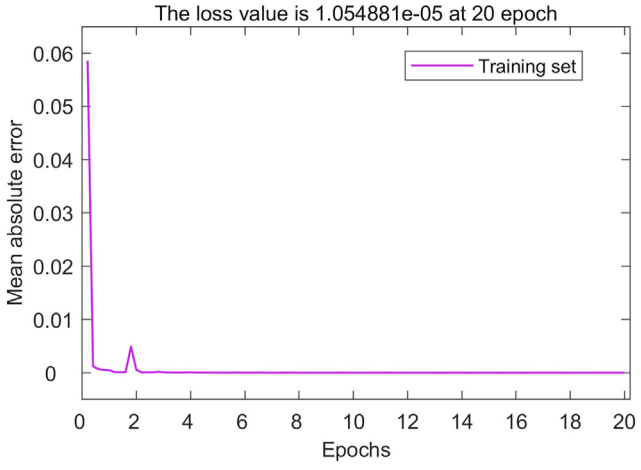


Fig. 7 – Error convergence curve.

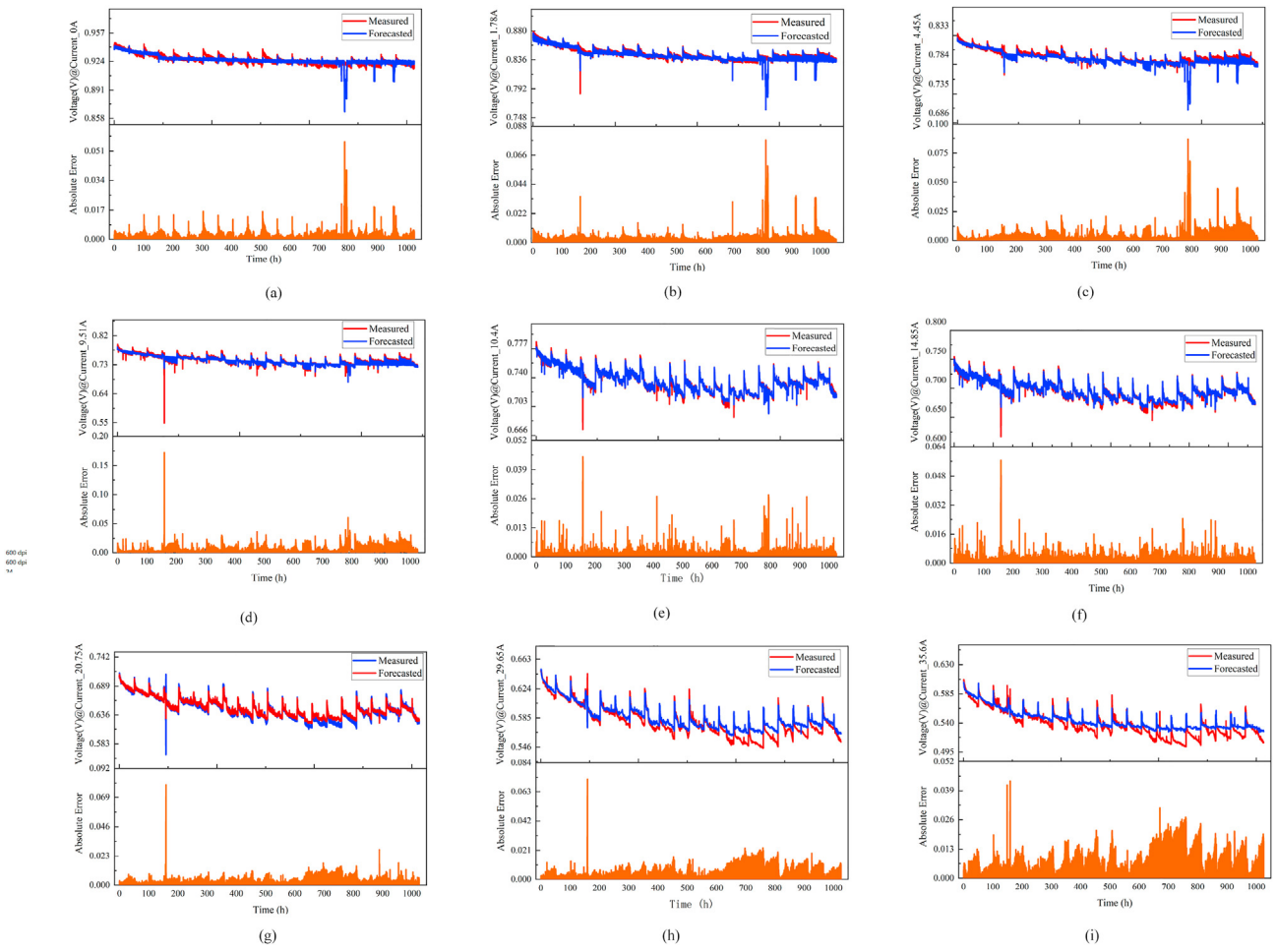


Fig. 8 – Comparison of measured values based on the voltages corresponding to the nine different current loading steps in the durability test cycle.

known covariate observed inputs. The illustration of forecasting with time-varying past known covariates and future known covariates is shown in Fig. 4.

The overall dynamic durability test is designed with reference to the European harmonized test protocols for Automotive PEM fuel cells [35]. It is composed of 3076 dynamic load cycles, lasting approximately 1008 h. According to the test protocols, the current of PEM fuel cells changes 35 times in per cycle. And it involves 9 different current loads (0%, 5%, 12.5%, 26.7%, 29.2%, 41.7%, 58.3%, 83.3%, 100 The current value is 35.6 A at 100% load. Every 304 cycles (about 100 h), the fuel cells system is shut down for 12 h and then restarted, resulting in performance recovery phenomenon.

Data preprocessing

High quality data achieves more accurate predictions. The original dataset contains too many data points and noise. This can lead to longer computation times and poorer prediction results. The answer to these problems is data preprocessing, which tracks degradation trends more accurately and efficiently. The first step is to sample the initial data. A dynamic load cycle that lasts 19 h and 41 s contains 35 current changes, so only 35 data points are selected in a cycle, which retain useful information in the signal. The data points selected are the fifth-to-last point in the interval with the same current value since the data in the middle is undisturbed. The length of the sampled dataset is 107,600 data points.

Then the data is checked for missing values and outliers. To better identify the time series, new time features hours and cycle position are added. The current of the fuel cell changes 35 times per test cycle. The cycle position is defined as the order of the 35 data points sorted by time within a cycle. It is a sequence of numbers starting from 1 to 35. The values of the raw data are scaled to the 0–1 range using a normalization method calculated as:

$$x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}} \tag{22}$$

where x_{min} and x_{max} represent the minimum and maximum values in each feature. The size of moving window and forecast horizon is set to five and one, respectively. The last step is to divide the dataset into a training set and a test set with a ratio of 6:4. The final processing result of hours and voltage data is shown in Fig. 5.

Table 5 – Calculation results of performance criteria for prediction model on the training set.

	Training Set				
	TFT	LSTM	LSTM-attention	GRU	GRU-attention
RMSE	0.003247	0.011066	0.008542	0.015879	0.013091
MSE	0.000010	0.000122	0.000073	0.000252	0.000171
MAE	0.002094	0.007549	0.006170	0.011751	0.009827
R^2	0.998602	0.984143	0.990593	0.967374	0.977808

Table 6 – Calculation results of performance criteria for prediction model on the testing set.

	Testing Set				
	TFT	LSTM	LSTM-attention	GRU	GRU-attention
RMSE	0.006721	0.017637	0.016409	0.018260	0.0159518
MSE	0.000045	0.000311	0.000269	0.000333	0.0002141
MAE	0.003568	0.013970	0.013002	0.014368	0.012016
R^2	0.991109	0.962568	0.967600	0.959877	0.971025

Model architecture

With the temporal fusion transformer (TFT) framework presented in the previous section, the model for the degradation prediction of PEM fuel cells is developed by training the TFT model. The training framework of this model used in predicting the degradation performance of PEM fuel cells is shown in Fig. 6. To implement the prediction model based on TFT, the major steps can be summarized as.

1. Data pre-processing and defining the prediction step,
2. Defining the structure of the prediction method,
3. Determining the hyperparameter by random searching,
4. Conducting the performance evaluation by using the trained model to predict the targets at each prediction step. The predicted results are compared with the measured values in the experiments and the errors between predicted results and measured values are calculated.

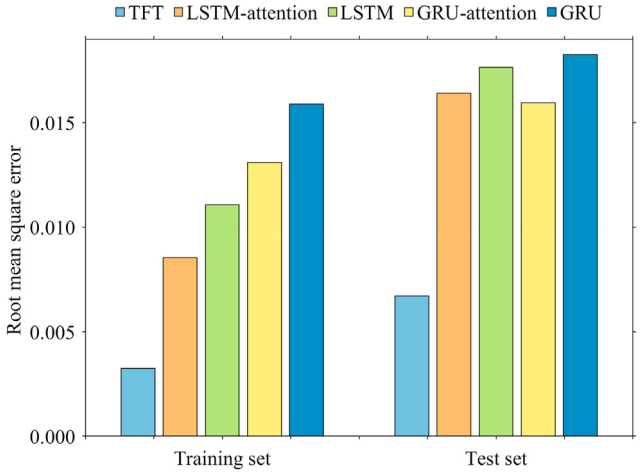


Fig. 9 – Comparison of root mean square error for different methods on the training set and test set.

Parameter definition of model

In this section, the model based on TFT is developed for predicting the degradation performance of PEM fuel cells. Seventeen parameters from the dataset are used as the model input covariates, as presented in Table 2 and Table 3. The covariates are used as a priori knowledge to assist in the prediction of the target series and are not directly involved in the prediction of the target. The different input covariates are classified into time-varying past known covariates and future known covariates according to the different types and heterogeneity of the input data. Table 2 lists the results of the statistical analyses of time-varying past known covariates of the model. Table 3 lists the results of statistical analyses as time-varying future known covariates of the model. The voltage is used as the model output. In this framework, hyperparameter optimization is applied to find the optimal significant parameters of the model. Due to the training efficiency and computational cost considerations considered, a random search is used to find the optimal values of the hyperparameter. The optimized ranges and optimal parameter values are listed in Table 4.

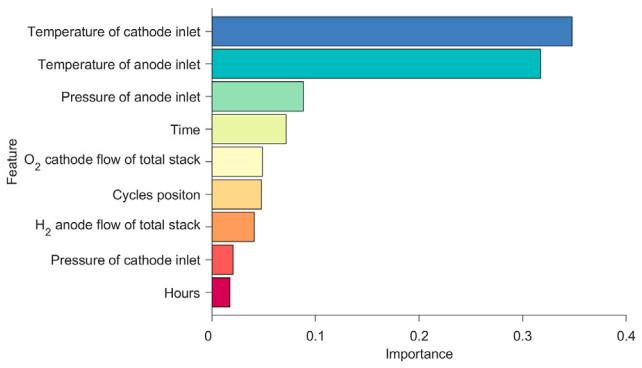


Fig. 10 – Importance of time-varying future known covariates.

Table 7 – Feature importance of time-varying future known covariates.

Variables	Importance
Temperature of cathode inlet	0.34763908
Temperature of anode inlet	0.31741962
Pressure of anode inlet	0.08823010
Time	0.07171467
O_2 cathode flow of total stack	0.04861942
Cycle position	0.04770441
H_2 anode flow of total stack	0.04085390
Pressure of cathode inlet	0.02058728
Hours	0.01723153

Table 8 – Importance of time-varying past known covariates.

Variables	Importance
Power	0.84844846
Pressure of cathode outlet	0.00833389
Temperature of anode outlet	0.00807252
Temperature of anode dewpoint water	0.00800865
Current	0.00753896
Temperature of cathode dewpoint water	0.00737205
Temperature of anode endplate	0.00722530
Voltage	0.00683582
Temperature of cathode outlet	0.00540717

Evaluation method

Four widespread time series forecasting evaluation metrics were utilized to estimate the performance of the presented model. Mean absolute error (MSE), root mean square error (RMSE), mean absolute percentage error (MAPE) and the coefficient of determination (R^2), are expressed as follows:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2 \tag{23}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2} \tag{24}$$

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{y_i - \hat{y}}{y_i}\right| \times 100 \tag{25}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y})^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2} \tag{26}$$

where N is the total number of samples used as input; $\hat{y}$ are the predicted results; y_i represents the measured value; $\bar{y}$ is the average value for the each sample y_i. The last phase of the development process involves evaluating the performance of the model. The results of the proposed model are compared with the measured data on the test set, and the prediction performance errors are calculated. The results of the prediction are further described in the next section.

Results and discussions

First, an interpretable data-driven method based on TFT and covariates for degradation prediction was developed by a specific training process. The degradation performance for both long-time and short-time periods was investigated and the predicted values were compared with the experimental measurements. Secondly, the performance of the method based on TFT is compared with four methods based on deep learning in the literature [17]: LSTM, GRU, LSTM-attention, and GUR-attention. Finally, model interpretability based on feature importance and multi-headed attention is further investigated.

Performance of the interpretable data-driven method

The error convergence curve of training the interpretable data-driven method is placed in Fig. 7. Since the dataset includes more than 1000 h of dynamic test data, it is challenging to depict in one figure the overall results of the interpretable data-driven method. The overall results are placed in Fig. 8, which is based on the voltages corresponding to the nine different current loading steps in the durability test cycle. From Fig. 8, the voltage measurements in the dynamic

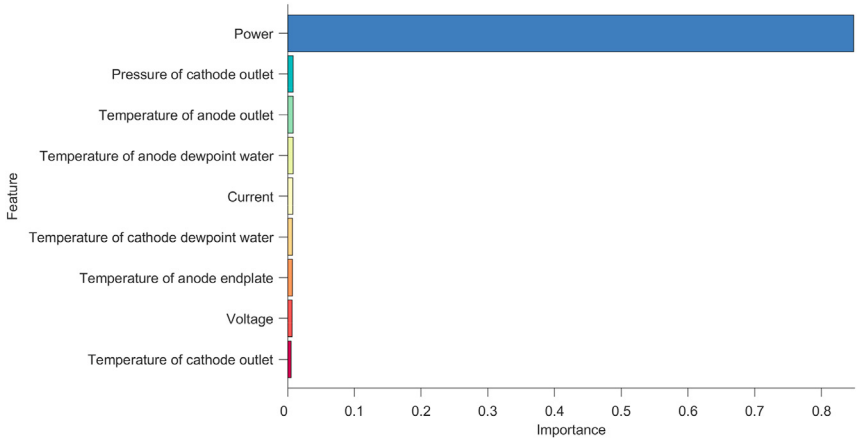


Fig. 11 – Importance of time-varying past known covariates.

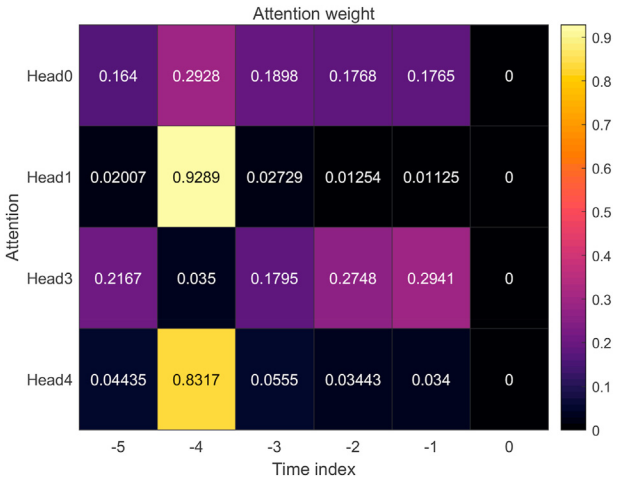


Fig. 12 – Multi-head attention weight based on time index.

durability tests of the PEM fuel cells were compared with the predicted output voltage.

Comparison of different models

The proposed method results on the training and test sets are compared with four methods obtained from the literature, as listed in Table 5 and Table 6, respectively. Zuo et al. [20] developed different prediction methods' long short-term memory (LSTM), gated recurrent units (GRUs), attention-based LSTM, and attention-based GRUs for predicting degradation performance of PEM fuel cells. Fig. 9 shows the comparison of the root mean square error for different methods on the training set and test sets. The interpretable data-driven method outperformed the other four methods on prediction performance.

Interpretability analysis

The variable selection network and multi-head attention network are the interpretable components for improving interpretability in model predictions. Therefore, the interpretative analysis can be further investigated from these two aspects. Fig. 10 shows the feature importance of time-varying future known covariates. The detailed values of time-varying future known covariates are listed in Table 7. The importance of both features, the cathode and anode inlet temperature, reached about 0.3. The degradation performance was directly correlated with the inlet and outlet temperatures and temperature cycling. This can be further demonstrated in the literature [36], where a change in thermal cycling leads to the degradation of membrane and catalytic layer properties. Fig. 11 shows the feature importance of time-varying past known covariates. The detailed values of time-varying past known covariates are listed in Table 8. The feature importance of power is reached 0.8, which is meaningful. It can be further explained in depth as the dynamic cycling variation of PEM fuel cells is affected by performance degradation since the load power varies. The catalytic platinum (Pt) layer and the carbon support are degraded during the power loading cycle. This could also be caused by the reduction of the carbon support surface area and the clustering of the catalyst nanoparticles, leading to the corrosion of the carbon support [37,38]. Fig. 12 shows the weight of the multi-head attention. The multi-headed attention mechanism focuses on both short-term and long-term time correlations but is more focused on long-term time correlations than on short -term time correlations.

Conclusion

The purpose of this study was to improve the prediction accuracy and credibility of PEM fuel cells degradation prediction results by fully considering the model interpretability. In addition, the experimental data has the potential to be exploited for further exploration. Heterogeneity of input features and the effect of significant covariates are taken into consideration. This data-driven method provides primary interpretable degradation predictions for temporal fusion transformers and covariates. The results is compared with four data-driven methods in recently published research. Then, the interpretability of the model is further investigated. The interpretable data-driven method for predicting the degradation of fuel cell voltage performance makes it possible to easily monitor and diagnose PEM fuel cells system in real time. This can be combined with a real-time control strategy to prolong PEM fuel cells life. The interpretability of the model makes the model more reliable and quantifies the importance of the covariates to provide a better guide for studying the degradation mechanism. Future work can be further extended to integrate with real-time control strategies for fuel cell systems to help improve the durability of fuel cells. The conclusions are summarized as follows.

(1) The method based on temporal fusion transformers achieved high accuracy in predicting the degradation performance of PEM fuel cells with root mean square errors of 0.0067. The proposed model could reduce the root mean square error by 58% compared to the attention-based gated recurrent unit.
(2) The interpretability results show that the cathode and anode inlet temperatures need special attention among the controllable features. The feature importance of both cathode and anode inlet temperatures is accounted for by 30%.
(3) Based on the interpretability analysis, model covariates have a direct effect on the prediction results. These include power, time, the pressure of the cathode inlet, and the pressure of the cathode outlet.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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